

Equations of Motion in SHO's

1. An object of mass 0.75 kg oscillates according to the equation

$$y(t) = 0.50 \cos(12.47 \cdot t) ,$$

where x is measured in meters and t is measured in seconds.

(a) What is the amplitude?

(b) What is the frequency?

(c) Plot the equation on Desmos (www.desmos.com/calculator) and explain how it the resulting graph matches your answers for (a) and (b).

(d) Calculate the spring constant k of this spring.

(e) At $t = 1.6$ s, what will be the *location* of the mass and what will be its *velocity*?

(f) At $t = 1.6$ s, what is the KE and PE of the system?

(g) What is the total energy of the system?

2. Now, try plotting $y = A \sin(2\pi f \cdot t + \phi)$ on Desmos. Use the “slider” capability of Desmos to investigate how changing f and A changes the graph.

Name: _____

Date: _____

Period: _____

Damped Harmonic Motion

